Plate Tectonics: Layers of the Earth

Earth is built in layers. The crust is the thin, rocky outer shell, about 5 to 70 kilometers thick. Beneath it lies the mantle, a thick layer of hot, slowly flowing rock that makes up most of the planet's volume. At the center is the core: a liquid outer core of iron and nickel surrounding a solid inner core.

The lithosphere combines the crust and the uppermost, rigid part of the mantle. It is broken into large pieces called tectonic plates. The plates float on the asthenosphere, a softer layer of the mantle that can flow over millions of years.

Plate Boundaries

Plates meet at three kinds of boundaries. At a divergent boundary, plates move apart and new crust forms, as at the Mid-Atlantic Ridge. At a convergent boundary, plates collide; one may sink beneath the other in a process called subduction, which builds volcanic arcs and deep ocean trenches. At a transform boundary, plates slide past each other horizontally, as along the San Andreas Fault.

Most earthquakes and volcanoes occur near plate boundaries. The Ring of Fire around the Pacific Ocean contains about three quarters of the world's active volcanoes.

Evidence for Plate Movement

Alfred Wegener proposed continental drift in 1912. He pointed to matching coastlines, identical fossils on separate continents, and similar rock formations across oceans. His idea was rejected until the 1960s, when seafloor spreading and magnetic stripes on the ocean floor explained how continents could move.

Today, GPS measurements show plates moving a few centimeters each year, about as fast as fingernails grow.